

Product Review Sentiment Analyzer & Aspect-Based NLP Platform

Project 29 | Enterprise Natural Language Processing & Customer Intelligence

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GitHub Repository: https://github.com/NAVYA1174/product-review-sentiment-analyzer	Technology Stack: Python, Scikit-Learn, NLTK, Streamlit, Plotly
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Executive Abstract

Abstract: In modern e-commerce ecosystems, customer reviews are essential drivers of consumer decisions and product strategy. However, conventional sentiment analysis methods often reduce feedback to coarse, binary classifications that obscure actionable insights. This project presents the Product Review Sentiment Analyzer (Project 29), an enterprise-grade Natural Language Processing (NLP) framework designed to extract fine-grained, dimension-specific intelligence from unstructured consumer feedback. The system incorporates a tri-engine hybrid ensemble combining rule-based heuristics (NLTK VADER), linguistic polarity and subjectivity modeling (TextBlob), and statistical machine learning (TF-IDF vectorization with L2-regularized Logistic Regression). Additionally, the platform integrates an Aspect-Based Sentiment Analysis (ABSA) engine that segments customer opinions across five vital product dimensions: Quality & Build, Performance, Price & Value, Design & Comfort, and Customer Support. To resolve nuanced consumer language, rule-based heuristics are deployed to detect sarcasm, negation flips, and emotional tone distributions. The model is deployed via an interactive Streamlit web dashboard featuring dynamic batch CSV ingestion, automated Net Sentiment Score (NSS) calculations, head-to-head competitor radar benchmarking, and automated executive intelligence reporting. Validated against multi-category product datasets with 100% test accuracy and passing unit test coverage, the framework bridges the gap between raw textual feedback and strategic product management.
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1. Problem Statement & Project Objectives

Commercial retailers and product managers frequently face information overload from thousands of user reviews. Simple star ratings and overall sentiment polarity fail to pinpoint specific product flaws (e.g., whether negative reviews stem from battery degradation, poor packaging, or inflated pricing). The core objectives of this project are:

- **Granular Dimension Extraction:** Isolate aspect-level feedback and compute Net Sentiment Scores (NSS) for targeted engineering action.
- **Hybrid Ensemble Inference:** Synthesize lexicon-based sentiment analysis with statistical machine learning to maximize recall across diverse vocabulary and slang.
- **Sarcasm and Emotion Awareness:** Mitigate false positives caused by ironic phrases and estimate emotional distribution (Joy, Trust, Frustration, Disappointment).

- **Production-Ready Deployment:** Deliver an interactive web dashboard with competitive product comparison, catalog filtering, and automated report exports.

2. System Architecture & Technical Methodology

The system follows a modular four-layer architecture spanning text preprocessing, multi-model sentiment inference, aspect extraction, and dashboard visualization:

Layer	Module / Component	Core Responsibility
Preprocessing	utils/text_cleaner.py	Contraction expansion (40+ patterns), emoji translation, negation-preserving stopwords filtering, and sentiment tokenization.
Sentiment Engine	models/sentiment_engine.py	Weighted consensus ensemble (VADER 40% + TextBlob 30% + ML 30%), sarcasm heuristic filters, and star rating conversion.
Aspect Mining (ABSA)	models/aspect_extractor.py	Sentence clause segmentation, taxonomy keyword matching across 5 pillars, domain-adapted scoring, and quote extraction.
Machine Learning	models/train_model.py	Sublinear TF-IDF feature extraction (1-2 n-grams, 5000 max features) paired with L2-regularized Logistic Regression.
Visualization & UI	app.py & utils/visualizer.py	Streamlit web interface featuring animated dial gauges, dual-polar radar charts, weekly trendlines, and CSV/PDF exporters.

3. Aspect-Based Sentiment Taxonomy & Metrics

To deliver actionable feedback, the engine monitors five universal e-commerce pillars using a domain-specific vocabulary:

Aspect Dimension	Monitored Semantic Signals	Business Impact
Quality & Build	durability, materials, sturdiness, broken, cracked, flimsy, peeling, hinges	Informs hardware QA, manufacturing tolerances, and supplier component reliability.
Performance	battery, speed, sound clarity, noise cancelling, sensors, lag, charging	Guides engineering firmware optimization, acoustic tuning, and battery power profiles.
Price & Value	worth, affordable, bargain, overpriced, expensive, rip-off, investment	Assists pricing strategy, discounting, and promotional bundle positioning.
Design & Comfort	ergonomic, lightweight, fit, heavy, sleek, blisters, padding, aesthetic	Guides industrial design adjustments, accessory sizing, and strap ergonomics.
Support & Delivery	customer service, warranty, refund, return policy, shipping speed, packaging	Optimizes post-purchase customer success SLAs and logistics fulfillment.

Net Sentiment Score (NSS) Formulation:

For each dimension, the Net Sentiment Score (NSS) is standardized between -100% and +100%:

- NSS = [(Positive Mentions - Negative Mentions) / Total Mentions] × 100%**
- *NSS > +20%*: Strong Competitive Advantage | *-10% ≤ NSS ≤ +20%*: Neutral Market Perception | *NSS < -10%*: Critical Churn Risk

4. Experimental Results & Model Evaluation

The machine learning classifier was trained and evaluated on stratified benchmark splits. The TF-IDF + Logistic Regression architecture demonstrated strong discriminative power across all sentiment classes:

Sentiment Class	Precision	Recall	F1-Score	Test Support
Negative	1.00	1.00	1.00	18
Neutral	1.00	1.00	1.00	10
Positive	1.00	1.00	1.00	17
Accuracy / Total	100%	100%	1.000	45

5. Verification & Unit Testing

A rigorous automated test suite was constructed under tests/test_analyzer.py to ensure end-to-end software engineering integrity. Key verified test cases include:

- **Text Normalization Test:** Validated emoji mapping and contraction resolution without stripping negation modifiers.
- **High Polarity Classification:** Verified positive scores (polarity > 0.20, predicted rating ≥ 4.0 stars).
- **Negative Feedback Classification:** Verified defect detection (polarity < -0.20, predicted rating ≤ 2.5 stars).

- **Sarcasm Detection Test:** Verified that ironic contradictory statements trigger the sarcasm heuristic and flip false-positive polarity.
- **ABSA Quote Isolation:** Verified that multi-clause compound sentences correctly route aspect mentions and sentiment labels.

Test Suite Result: 5 tests executed, 0 failures, 0 errors (Status: OK).

6. Conclusion & Future Roadmap

The Product Review Sentiment Analyzer successfully demonstrates how modern NLP methodologies can be operationalized into a high-utility decision support system. By combining multi-engine ensembles, aspect mining, and an interactive exploratory dashboard, the application translates ambiguous unstructured text into precise product roadmaps. Future enhancements include fine-tuning domain-specific transformer models (such as DistilBERT or RoBERTa) and implementing direct live scraping integrations for Amazon, Flipkart, and Shopify APIs.

Official Project Submission | Prepared for Review | Repository:
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